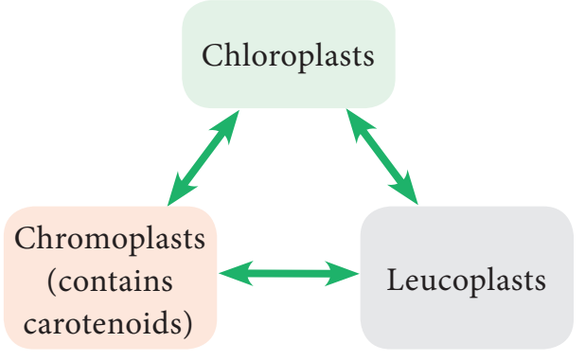
| Plastids | |
|---|---|
| **Chromoplasts** | **Leucoplasts** |
| Coloured Plastids | Colourless plastids store food materials |
| **Chloroplast**<br>Occurs in green algae and higher plants.<br>Pigments chlorophyll *a* and *b* | **Amyloplast** – stores – starch |
| **Phaeoplast**<br>Brown algae and dinoflagellates.<br>Pigment-fucoxanthin | **Elaioplast** – store – lipids (oils).<br>Seed of monocot and dicots. |
| **Rhodoplast**<br>Red algae<br>Pigment-Phycoerythrin | **Aleuroplast** (or) **Proteoplast**<br>Stores – Protein |

According to Schimper, different kind of plastids can transform into one another.




### 6.6.6 Chloroplast

Chloroplasts are vital organelle found in green plants. Chloroplast has a double membrane the outer membrane and the inner membrane separated by a space called **periplastidial space**. The space enclosed by the inner membrane of chloroplast is filled with gelatinous matrix, lipo-proteinaceous fluid called **stroma**. Inside the stroma there are flat interconnected sacs called **thylakoid**. The membrane of thylakoid enclose a space called **thylakoid lumen**.

**Grana** (singular: Granum) are formed when many of these thylakoids are stacked together like pile of coins. Light is absorbed and converted into chemical energy in the granum, which is used in stroma to prepare carbohydrates. Thylakoid contain chlorophyll pigments. The chloroplast contains osmophilic granules, 70s ribosomes, DNA (circular and non histone) and RNA. These chloroplast genome encodes approximately 30 proteins involved in photosynthesis including the components of photosystem I & II, cytochrome bf complex and ATP synthase. One of the subunits of RuBisco is encoded by chloroplast DNA. It is the major protein component of chloroplast stroma, single most abundant protein on earth. The thylakoid contain small, rounded photosynthetic units called **quantosomes**. Chloroplast is a semi-autonomous organelle and divides by fission (Figure 6.16).

**Functions:**

- Photosynthesis
- Light reactions takes place in granum,
- Dark reactions take place in stroma,
- Chloroplast is involved in photo-respiration.

### 6.6.7 Ribosome

Ribosomes were first observed by **George Palade** (1953) as dense particles or granules in the electron microscope. Electron microscopic observation reveals that ribosomes are composed of two rounded sub units, united together to form a complete unit. $Mg^{2+}$ is required for structural cohesion of ribosomes. Biogenesis of ribosome is a *de nova* formation, auto replication and nucleolar origin. Each ribosome is made up of one small and one large sub-unit Ribosomes are the sites of protein synthesis in the cell. Ribosome is not a membrane bound organelle (Figure 6.17).



11th Botany_EM_Chapter 6.indd 156 05-01-2023 18:25:12